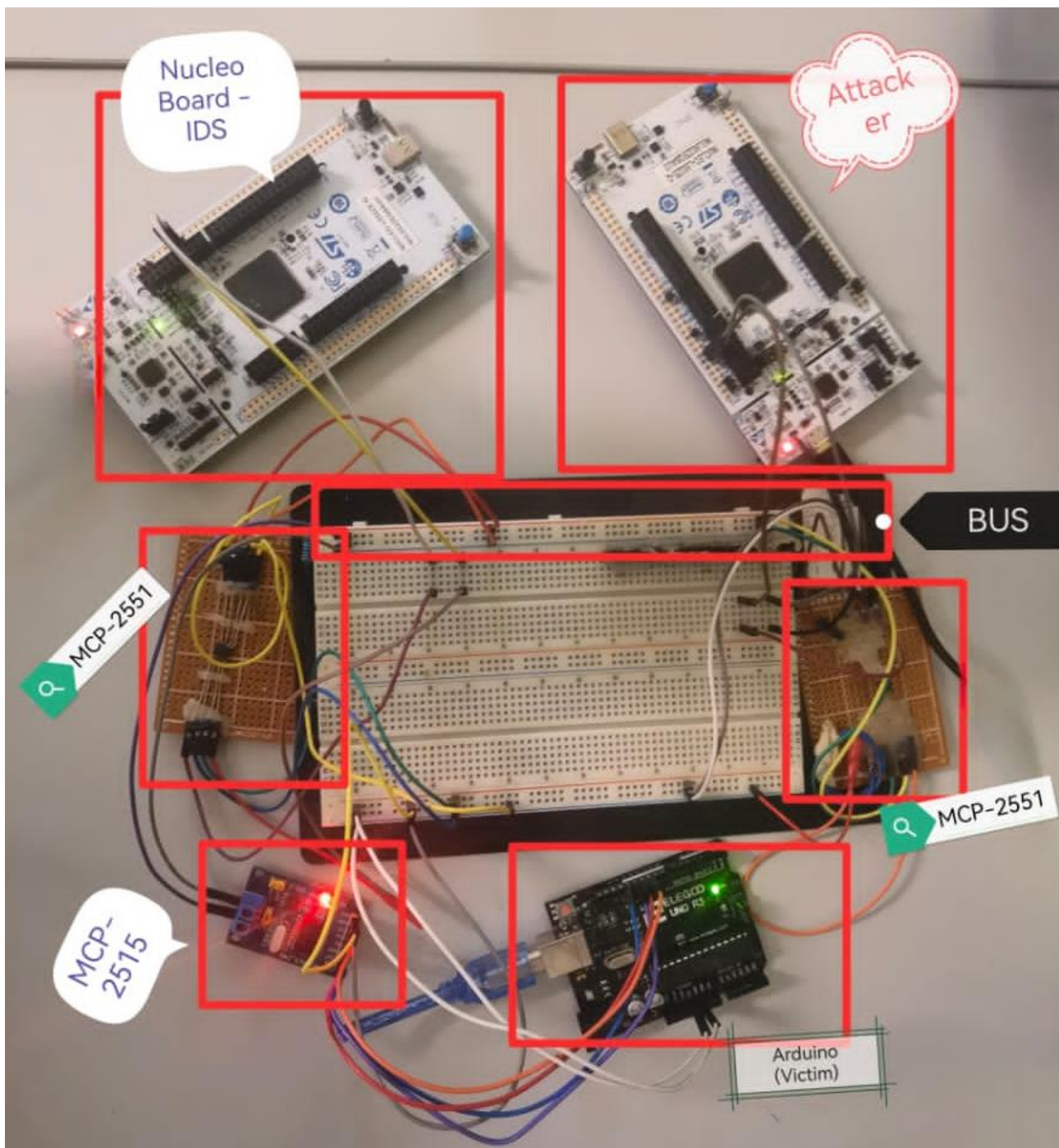
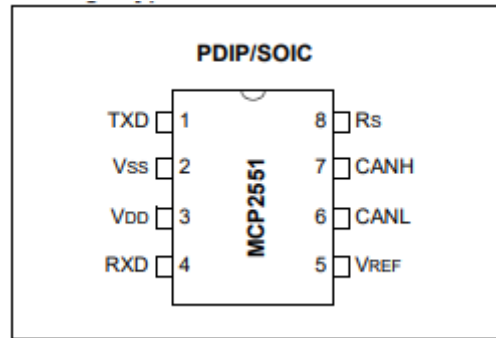


## EXPERIMENTAL SETUP



The above picture shows the entire experimental setup. First of all, I am going to explain the cabling of Victim which is implemented on **Arduino board**. 5v power and ground pins of Arduino board are connected to bread board to provide power and ground connections to MCP-2515 CAN controller and MCP-2551 CAN transceivers. Then the pin 10 which is CS pin is connected with CS pin of MCP-2515, pin 11 of Arduino board is connected to SI pin (which is MOSI), pin 12 is connected to SO (which is MISO) of MCP-2515 and finally pin 13 of Arduino board is connected to SCK pin of MCP-2515. Then CANH (purple wire) and CANL (grey wire) pin of MCP-2515 are attached to the bread board so that I can connect the same of other 2, MCP-2551 CAN transceivers to form a CAN bus.

Now coming over to Nucleo board, both attacker board and IDS board have same cabling so I will be explaining that of IDS. Pin PA6 (yellow wire) which is MISO is connected to RXD pin of MCP-2551 and pin PA7 (white wire) which is MOSI is connected to TXD pin of MCP-2551. VSS pin of MCP-2551 is connected to ground of bread board, VDD which is power line is connected to 5V power line of bread board (which is getting power/ ground connection from Arduino). CANH (red wire) is connected with CANH of Arduino and CANL (orange wire) is connected with CANL of the same. RS pin or MCP-2551 is connected to ground as well.



### MCP-2551 PIN Layout

These same connections are made for Nucleo board used as attacker as well. Pin PA6 (brown wire) which is MISO is connected to RXD pin of MCP-2551 and pin PA7 (grey wire) which is MOSI is connected to TXD pin of MCP-2551. VSS pin of MCP-2551 is connected to ground of bread board, VDD which is power line is connected to 5V power line of bread board (which is getting power/ ground connection from Arduino). CANH (yellow wire) is connected with CANH of Arduino and CANL (green wire) is connected with CANL of the same. RS pin or MCP-2551 is connected to ground as well.